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## Interleukin-2-induced skin inflammation

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### Abstract

Recombinant human interleukin-2 (IL-2) has been used to treat inflammatory diseases and cancer, however, side effects like skin rashes limit the use of this therapeutic. To identify key molecules and cells inducing the side effect, we characterized IL-2-induced cutaneous immune reactions and investigated the relevance of CD25 (IL-2 receptor  $\alpha$ ) in the process.

We injected IL-2 intradermally into wild type mice and observed increases of immune cell subsets in the skin with preferential increases in frequencies of IL-4- and IL-13-producing group 2 innate lymphoid cells (ILC2s) and IL-17-producing dermal  $\gamma\delta$  T cells. This overall led to a shift towards type 2/type 17 immune responses. In addition, using a novel topical genetic deletion approach, we reduced CD25 skin-specifically on all cutaneous cells and found that IL-2-dependent effects were reduced, hinting that CD25 – at least partly – induces this skin inflammation. Reduction of CD25 specifically on skin Tregs further augmented IL-2-induced immune cell infiltration, hinting that CD25 on skin Tregs is crucial to restrain IL-2-induced inflammation. Overall, our data support that innate lymphoid immune cells are key cells inducing side effects during IL-2 therapy and underline the significance of CD25 in this process.

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CS, JNC, and MDR conceived and designed research; CS and JNC performed experiments. CS, JNC, and DS analyzed data; CS, JNC, SD, VN, DS, AB, KS, and MDR interpreted results of experiments; CS prepared figures and drafted manuscript; all authors read and approved the final manuscript.

Conflict of interest

The authors declare that they have no competing interests.

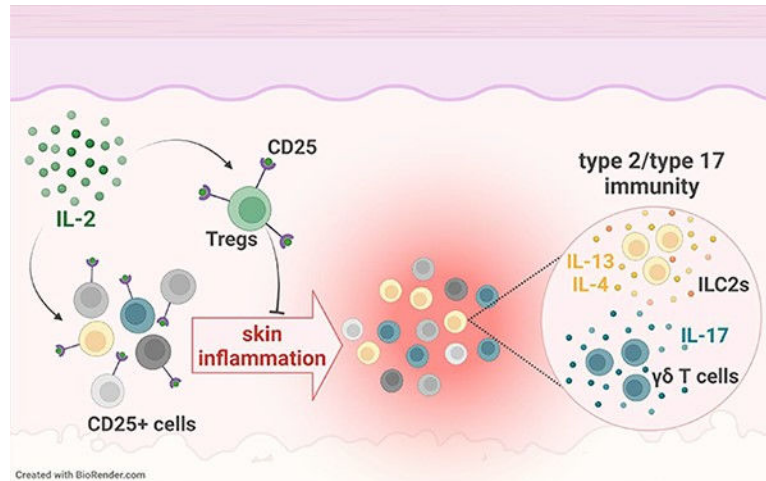
Ethics approval statement

Animal experiments were performed in accordance with guidelines established by the Institutional Animal Care and Use Committee and Laboratory Animal Resource Center of the University of California, San Francisco.

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## Graphical Abstract



## Keywords

Immunotherapy; IL-2 therapy; drug-induced skin inflammation; innate lymphoid cells;  $\gamma\delta$  T cells

## Introduction

As one of the first immunomodulatory cancer therapies to be approved by the FDA, recombinant interleukin-2 (IL-2; aldesleukin), aimed to induce therapeutic effects by activating T and natural killer (NK) cells. However, high-dose application necessary for anti-tumor activity led to severe systemic side effects, including skin rashes in up to 72% of the patients (1). Currently, modifications of IL-2 have been undertaken to increase specificity of treatment for both cancer and autoimmune diseases (2, 3). However, a better understanding of the molecular mechanisms of IL-2-induced dermatitis is needed.

Affinities to IL-2 are determined by differential expression of IL-2 receptor (IL-2R) complexes on the respective cell. The dimeric intermediate-affinity IL-2R $\beta\gamma$  (CD122 and CD132) is shared by IL-15 (4) and expressed by resting T and NK cells, among others, which can in turn be activated at high IL-2 concentrations (5–7). Additional expression of the IL-2R $\alpha$  subunit (CD25), which is specific for IL-2, can induce formation of the high-affinity trimeric IL-2R $\alpha\beta\gamma$  which can bind IL-2 at picomolar concentrations (8). Regulatory T cells (Tregs) which have high CD25 expression (5, 9) are able to scavenge free IL-2 through their high-affinity IL-2R, further promoting immune homeostasis (10, 11). In addition to Tregs, innate lymphoid cells (ILCs) including activated NK cells as members of group 1 ILCs (ILC1s) (12) as well as activated  $\alpha\beta$  and  $\gamma\delta$  T cells might express the high-affinity IL-2R (6, 7, 13–15).

Similar to patients treated with high-dose IL-2, dermatitis occurred in Rag<sup>-/-</sup> mice treated with IL-2 and a CD25-specific antibody (13), suggesting a central role of CD25 expression in IL-2-induced skin reactions. Histologically, skin reactions showed infiltrates of eosinophils and neutrophils in both mice and humans (13, 16, 17). In addition, expansion

of group 2 ILCs (ILC2s), which are the innate counterpart of type 2 T helper cells, was observed with increased production of IL-5 and IL-13 upon IL-2 treatment (13, 18). Together, these data led us to hypothesize that skin rashes during IL-2 therapies are CD25-dependent and characterized by type 2 immunity.

To test these hypotheses, we applied recombinant human IL-2 (aldesleukin) intradermally in mice and characterized the resulting immune response. Second, to evaluate the role of CD25 in IL-2-induced dermatitis, we deleted CD25 locally on skin cells and measured the impact of CD25 reduction on IL-2-induced skin reactions. Lastly, because CD25-expressing Tregs have a central role in maintaining immune homeostasis, we analyzed the effects of CD25 deletion on skin Tregs upon IL-2 injections.

## Results

### IL-2-induced skin inflammation is characterized by increases in innate immune cells

To characterize cutaneous immune reactions elicited by IL-2, wild type mice were injected intradermally with recombinant human IL-2. For dose-finding experiments, IL-2 was injected once or twice daily for 7 consecutive days (Figure 1A). The IL-2 dose applied here roughly corresponds to one high-dose intravenous injection given in the clinics (19). However, the intermediate IL-2R $\beta\gamma$  in mice has a lower affinity to human IL-2 than human IL-2R $\beta\gamma$  (20, 21) and, thus, the chosen dosing regimen aimed to induce local over systemic effects in mice.

Histological analysis of skin at the injection site revealed mild inflammatory cell infiltration in control-treated skin (Figure 1B). In contrast, injections of IL-2 once per day induced prominent inflammatory cell infiltrates consisting of lymphocytes, granulocytes, and macrophages (Figure 1C). Numbers of these cells were highly increased in skin of mice injected twice daily (Supplementary Figure S2), leading to cell infiltrates within large subdermal areas. To ensure only local immune reactions in skin without inducing systemic effects, IL-2 injections were performed once daily in all further experiments.

Flow cytometric analysis of immune cell populations in skin and skin-draining lymph nodes (SDLN) supported histological findings with significant increases in numbers of CD45<sup>+</sup> cells in the side of skin treated with IL-2 as compared to control sides (Figure 2B). These increases in skin immune cell numbers were observed for CD4<sup>+</sup> and CD8<sup>+</sup> T cell subsets (5.4- and 4.1-fold, respectively), dermal  $\gamma\delta$  T cells (8.6-fold), ILCs (9.0-fold), and CD3-NK1.1<sup>+</sup> “NK-like” group 1 ILCs (further referred to as ILC1s) (27.2-fold), but not for dendritic epidermal T cells (DETCs) (Figure 2C) and were accompanied by induction of the proliferation marker Ki-67 in T cell subsets and ILCs (Supplementary Figure S3A). In addition to increases in lymphocytes, IL-2 injections induced increases of neutrophil (13.6-fold) and eosinophil numbers (7.3-fold) in skin (Figure 2D). Notably, frequencies of immune cell subsets were only significantly increased for dermal  $\gamma\delta$  T cells (mean 4.4% to 7.3%), ILCs (mean 4.9% to 8.0%), and ILC1s (mean 2.1% to 10.5%) (Figure 2E), but not for CD4<sup>+</sup> or CD8<sup>+</sup> T cell subsets or granulocytes (Supplementary Figure S3B).

CD25 as the subunit crucial to form the high-affinity trimeric IL-2R could be central in inducing IL-2-dependent side effects. Upon IL-2 injections, slight increases in frequencies of CD25<sup>+</sup> cells were observed for CD4<sup>+</sup> Teffs and CD8<sup>+</sup> T cells (Figure 2F) in addition to increases in mean fluorescence intensity (MFI) of CD25 on CD4<sup>+</sup> Teffs (Supplementary Figure S3C). Strikingly, CD25 expression of dermal  $\gamma\delta$  T cells, which only showed low if any CD25 expression under control conditions, was significantly increased upon IL-2 treatment, both regarding CD25<sup>+</sup> frequencies (mean 2.0% vs. 13.6% upon IL-2 injection) as well as MFI (Figure 2F; Supplementary Figure S3C).

### Skin-specific reduction of CD25 reduces IL-2-induced accumulation of cells in the skin

To clarify the role of CD25-expressing cells in IL-2-induced skin inflammation, we aimed to delete CD25 specifically on skin cells by applying a newly developed method which allows inducible skin-specific, non-systemic deletion of target genes (22). Rosa26<sup>CreERT2</sup>xCD25<sup>fl/fl</sup> mice were generated and 4-Hydroxytamoxifen (4-OHT) was applied topically to induce pan-cellular CD25 deletion on one side of the back skin of these mice (Supplementary Figure S4A). Acetone served as a vehicle control on the contralateral back skin, enabling us to investigate changes within the same mouse. Upon 4-OHT application without IL-2 injections, CD25 expression of skin Teffs, ILCs, and type 1 conventional DCs was almost completely abolished with frequencies of CD25<sup>+</sup> cells <3% (Supplementary Figure S4B), which was reflected by reduced MFIs on the respective cells (Supplementary Figure S4C). In addition, CD25 expression on skin Tregs was significantly reduced with respect to CD25<sup>+</sup> cell frequencies (mean 11.9%) and total MFI. No effects on CD25<sup>+</sup> cell frequencies were observed in SDLN, providing evidence that deletion of the receptor was restricted to skin (Supplementary Figure S4D).

To investigate effects of skin-specific CD25 deletion on IL-2-induced skin inflammation, 4-OHT was applied to induce deletion prior to initiation of IL-2 injections. Application of 4-OHT was continued during IL-2 treatment to ensure continuous CD25 deletion (Figure 3A). The resulting reduction of CD25 in skin of IL-2-treated mice was partial but nonetheless significant for Tregs, DETCs, ILCs including ILC1s, and type 1 conventional DCs as shown by frequencies of CD25<sup>+</sup> cells and MFIs (Figure 3B+C). Dermal  $\gamma\delta$  T cells showed slight decreases in MFI of CD25. Lower deletion efficacy upon IL-2 injection might be due to preferential proliferation of cells which remain CD25<sup>+</sup>. Furthermore, 4-OHT was found to mostly penetrate upper layers of the skin including epidermis and dermis (unpublished observations). Therefore, IL-2-induced cell infiltrates in lower sections such as the subcutis will be less affected by 4-OHT, resulting in incomplete deletion. In contrast to the skin, frequencies of CD25<sup>+</sup> immune cells in corresponding SDLN were largely similar between control and 4-OHT-treated sides (Supplementary Figure S5).

Upon reduction of CD25 expression on the 4-OHT-treated side, total numbers of immune cells were significantly decreased in skin (Figure 3D), with reduced Teffs, Tregs, dermal  $\gamma\delta$  T cells, and ILCs including ILC1s (Figure 3E) as well as eosinophils but not neutrophils (Figure 3F). Numbers of CD8<sup>+</sup> T cells, on the other hand, were unaltered – comparable to CD25 expression on CD8<sup>+</sup> T cells on the 4-OHT-treated vs. control side (Figure 3B+D). Proportions of T-bet<sup>+</sup> type 1 and GATA-3<sup>+</sup> type 2 T helper cells and ILCs were unaltered

upon CD25 reduction (Figure 3G). Together, these data suggest that CD25 expression on skin immune cells is required for their accumulation following intradermal IL-2 injection.

### CD25 reduction on Tregs increases IL-2-induced accumulation of cells in skin

Tregs are known to exert part of their immunoregulatory role through expression of CD25 (23, 24). To investigate the impact of CD25 expression on Tregs in controlling IL-2-induced inflammation in the skin, we aimed to specifically delete CD25 on skin Tregs using FoxP3<sup>CreERT2</sup>xCD25<sup>f/fl</sup> mice and the topical 4-OHT treatment approach (Figure 4A). Effects of skin-specific CD25 deletion on FoxP3<sup>+</sup> Tregs without IL-2 injections were extensively described elsewhere (22). Intradermal IL-2 injections and continuous 4-OHT treatment (Figure 4A) led to significantly reduced frequencies of CD25<sup>+</sup> Tregs to an average of 58.7% (Figure 4B). Remaining CD25<sup>+</sup> Tregs displayed reduced CD25 expression (Figure 4C). On the other hand, while CD25 expression was not affected on most cells, expression on ILC1s and type 1 conventional DCs was diminished 1.5-fold and 2.2.-fold, respectively (Figure 4B). Frequencies of CD25<sup>+</sup> cells in SDLN were found to be slightly decreased on Tregs, ILC1s, and type 1 conventional DCs (Supplementary Figure S6).

Strikingly, numbers of immune cells significantly increased (3-fold) in 4-OHT-treated skin upon IL-2 injections whereas numbers in corresponding lymph nodes were similar for both control and 4-OHT-treated sides (Figure 4D). Similarly, absolute numbers of all T cell subsets – except DETCs –, ILCs including ILC1s, and granulocytes were highly increased in 4-OHT-treated skin (Figure 4E+F). In addition, FoxP3 expression in Tregs was slightly reduced in skin treated with 4-OHT (Figure 4G) and a shift of the Teff:Treg ratio was observed towards Teffs (Figure 4H). Significant decreases of type 1 and 2 T helper cell (mean 23.4% vs. 4.9% and 27.4% vs. 17.0%, respectively) as well as ILC1 frequencies (mean 50.1% vs. 23.6%) were observed upon CD25 reduction on Tregs (Figure 3I). Proportions of ILC2s were unaltered (Figure 3I). Taken together, these data indicate that CD25 expression on skin Tregs restrains accumulation of cutaneous immune cells resulting in a higher Teff:Treg ratio following intradermal IL-2 injections.

### Dermal $\gamma\delta$ T cells and ILC2s induce mixed type 2/type 17 immunity in skin of IL-2-injected mice

Intradermal IL-2 injections induced increases in various immune cell numbers at the injection sides with specific increases in  $\gamma\delta$  T cells, and ILCs, including ILC1s. To further characterize the quality of the immune response induced by IL-2 treatment, intracellular cytokine expression was investigated in IL-2-injected wild type mice using flow cytometry (Figure 5A). While prevalence of interferon- $\gamma$  (IFN- $\gamma$ )<sup>+</sup> immune cells was significantly decreased (9.6% vs. 6.6% of CD45<sup>+</sup> cells), frequencies of IL-17<sup>+</sup> CD45<sup>+</sup> cells were markedly enhanced in response to IL-2 injections (5.2% vs. 9.1%) (Figure 5B). Frequencies of tumor necrosis factor  $\alpha$  (TNF- $\alpha$ )<sup>+</sup>, IL-4<sup>+</sup>, and IL-13-producing CD45<sup>+</sup> immune cells were largely unaltered. Overall, IL-2 injections led to a shift towards type 2/type 17 immune responses compared to control conditions, while type 1 immune responses were unaltered or slightly decreased (Figure 5C).

To determine cells responsible for shifts in immunity, intracellular cytokine production was further analyzed in immune cell subpopulations. While no significant changes in IFN- $\gamma$ -producing Teffs and CD8<sup>+</sup> T cells were observed, prevalence of IFN- $\gamma$ <sup>+</sup> ILC1s was significantly decreased (Figure 5D) suggesting that ILC1s may become dysfunctional due to overstimulation from daily IL-2 application, similar to tumor-infiltrating NK cells in mice and humans (25, 26). Significant increases in type 2 cytokine-producing ILCs, namely group 2 ILCs, were found upon IL-2 treatment (1.0% vs. 3.0% for IL-4-producing cells; 4.7% vs. 11.3% for IL-13-producing cells). The most striking changes in cytokine production were observed for dermal  $\gamma\delta$  T cells which showed significant and prominent increases in frequencies of IL-17-producing cells from an average of 29.2% to 56.1% (Figure 5D). In conclusion, we observed a mixed type 2/type 17 immunity in response to IL-2 injections with increases in frequencies of IL-4- and IL-13-producing ILC2s and IL-17-producing  $\gamma\delta$  T cells.

## Discussion

High-dose IL-2 therapy has been a promising treatment option for malignant cancers but was limited due to systemic toxicity. Still, pathophysiological mechanisms of IL-2-induced side effects such as skin rashes are obscure. Here, we provide new molecular and cellular insights of IL-2-induced skin inflammation and relevance of CD25 using different mouse models.

Intradermal IL-2 injections in wild type mice led to striking increases in several lymphoid immune cell numbers around the injection site. This might be due to infiltration of cells towards IL-2 or other inflammatory stimuli such as described for T cells (27, 28) or in inflamed tissue for ILC2s and  $\gamma\delta$  T cells (29, 30). Besides recruitment of cells, IL-2 can directly induce proliferation of cutaneous T cells and ILC2s in mice and humans (13, 31, 32). Therefore, increasing Ki-67 expression in cells of IL-2-treated skin could hint towards local proliferation of tissue-resident as well as migrated cells. We further observed infiltrates of neutrophils and eosinophils in response to IL-2 injections. Similar observations of lymphoid and granulocytic infiltrates were made after subcutaneously injecting IL-2 into rats (33) and in early clinical reports of high-dose IL-2-treated patients suffering from skin rash (16, 17, 34, 35). In our current study, we observed increases in the representation of ILCs including ILC1s and dermal  $\gamma\delta$  T cells, accompanied by significant changes in cytokine production in all three cell populations which hints towards a more specific effect of IL-2 on these innate cell subsets.

Frequencies of IL-4- and IL-13-producing ILCs were increased upon IL-2 injections – signature cytokines for group 2 ILCs (12). Both human as well as murine ILC2s are well described to be activated in response to IL-2, leading to secretion of type 2 cytokines (13, 18, 32). In this context, ILC2s seem to play a central role in IL-2-induced eosinophilia (18) and skin rashes in mice treated with the antibody JES6-1 and IL-2 (13). Our data further support the central role of ILC2s in IL-2-induced side effects as ILC2s seem to be responsible for the shift towards type 2 immunity observed in response to IL-2 injections. Furthermore, we found that IL-2 injections highly increased prevalence of IL-17<sup>+</sup> dermal  $\gamma\delta$  T cells. In mice, IL-17 production of  $\gamma\delta$  T cells is induced by IL-2 alone without further stimuli (36), hinting that IL-2 injections might directly induce IL-17 production in our study.



Furthermore, Shibata et al. reported that CD25+  $\gamma\delta$  T cells produce IL-17 and that CD25 expression was dependent on IL-2 (37). Therefore, upregulation of CD25 on  $\gamma\delta$  T cells might be crucial to induce IL-17 production observed in our study, suggesting a central role of CD25 expression in inducing the shift towards type 17 immune reactions in response to IL-2 injections. Besides direct IL-2-induced effects, a variety of cytokines are secreted during IL-2 therapy in mice and humans (18, 35, 38), which might further impact cutaneous cells indirectly. For instance, cell plasticity in the skin of IL-2-treated mice could be affected by the complex inflammatory milieu induced by IL-2 injections. Changes in plasticity have been reported for ILCs during the onset of psoriasis (39) or  $\gamma\delta$  T cells stimulated with IL-23, IL-1 $\beta$ , and IL-2 (40). Similarly, possible secretion of cytokines such as IL-33 in response to tissue damage could further support IL-2-induced type 2 responses by directly activating type 2 T helper cells or ILC2s, leading to synergistic effects of IL-33 and IL-2 on type 2 cytokine secretion (41–43). The overall shift towards a mixed type 2/type 17 immunity along with mixed neutrophilic and eosinophilic infiltrates into IL-2-treated skin might resemble pathologies such as atopic dermatitis or psoriasis (44, 45).

To further investigate the relevance of CD25 expression during IL-2-induced skin inflammation, we aimed to specifically delete CD25 on all skin cells. Although we achieved moderate partial rather than full deletion, we found that immune cell numbers were reduced upon IL-2 injections. In previous studies, macroscopic skin rashes in mice were observed when combining IL-2 treatment with the CD25-specific antibody JES6–1 (13). Furthermore, modifying IL-2 therapy to reduce or prevent binding to CD25+ cells reduces the occurrence of IL-2-induced side effects such as systemic inflammation or eosinophilia in preclinical models (46, 47). Together with our data, these observations highlight the role of the IL-2R $\alpha$  subunit in inducing side effects during IL-2 therapy.

In contrast to CD25 reduction on all skin cells, skin-specific reduction of CD25 on FoxP3+ Tregs drastically increased inflammatory infiltrates upon IL-2 injections. These observations were accompanied by an imbalance in the Teff:Treg ratio in favor of CD4+ Teffs, highlighting the role of CD25 on Tregs to maintain immune tolerance (23, 24). Furthermore, FoxP3 expression was slightly decreased upon reduction of CD25 on Tregs and IL-2 treatment, hinting towards a reduction of regulatory function (48, 49). In a previous study, we found that skin-specific CD25 deletion in FoxP3+ Tregs did not affect FoxP3 expression in skin Tregs under steady-state conditions (22). Therefore, decreased FoxP3 expression in the current study seems to be due to the additional IL-2 stimulation and could be an indirect effect due to IL-2-induced inflammation in the skin of mice (48, 49). Furthermore, decreases in type 1 and 2 T helper cells as well as ILC1s hint towards a higher prevalence of ROR $\gamma$ t-expressing cells, suggesting a shift to type 17 immunity upon CD25 reduction on Tregs. Overall, our data highlight the crucial role of CD25 on Tregs on restraining IL-2-induced inflammation in the skin.

Together, we show here that IL-2 injections induce cutaneous immune reactions characterized by a shift towards type 2/type 17 immunity which largely seems to be induced by ILC2s and dermal  $\gamma\delta$  T cells. In addition, while CD25-expressing Tregs suppress these inflammatory reactions, reduction of CD25 on all skin cells reduced IL-2-induced inflammation, showing that the receptor is responsible in mediating at least part

of these inflammatory reactions. Our study gives new molecular and cellular insight into IL-2-dependent skin reactions, identifying a central role of innate immune cells in inducing the side effect. The results presented here might therefore help to increase safety of current or future IL-2-based therapeutics.

### Data Limitations and Perspectives

In this study, we demonstrate that IL-2 induces skin inflammation as evidenced by increases in immune cell numbers and a shift towards type 2/type 17 immunity. To differentiate effects induced by IL-2 directly from indirect effects induced by produced cytokines, mediators such as IL-13, IL-17, or IL-33 might be reduced by blocking antibodies or in knockout mouse models. Similarly, deleting ILCs or  $\gamma\delta$  T cells – cells which seem to be responsible for the shift in immunity – could help to further understand the role of these unconventional immune cells in inducing side effects during IL-2 therapy.

## Methods

### Mice

All animals used for experiments were held in specific pathogen-free cages in the animal facility of the University of California, San Francisco. Mice of both genders were included, and littermate controls were used. Genetic strains of mice included wild type C57BL/6 (B6), Rosa26-CreERT2, FoxP3-CreERT2, and CD25<sup>fl/fl</sup> (Il2ra<sup>tm1c(EUCOMM)Wtsi</sup>) (all on B6 background). Animal experiments were performed in accordance with guidelines established by the Institutional Animal Care and Use Committee and Laboratory Animal Resource Center of the University of California, San Francisco.

### Administration of 4-Hydroxytamoxifen and IL-2

Dorsal back hair of mice was shaved along both flanks leaving a stripe of hair down the middle. To induce Cre-mediated recombination, 100  $\mu$ L of 4-Hydroxytamoxifen (4-OHT) (50  $\mu$ g/mL, Sigma-Aldrich; diluted in acetone, Sigma-Aldrich) was applied dropwise to the surface of one side of shaved back for 5 consecutive days starting at day -7 before IL-2 injections (22). As a control, 100  $\mu$ L acetone was applied to contralateral shaved back skin. For IL-2 injections, human recombinant IL-2 (aldesleukin) was diluted in PBS and 30  $\mu$ g/kg/injection in 100  $\mu$ L was intradermally injected (approx. 10,000 IU/injection), either to both sides of shaved back skin or unilaterally in experiments using wild type B6 mice. In experiments using B6 mice, 100  $\mu$ L PBS was injected intradermally in contralateral back skin as a control. IL-2 was intradermally injected for 7 consecutive days and 4-OHT or acetone control were topically applied during IL-2 injections daily for 7 days.

### Tissue processing

After euthanasia of mice, skin and skin-draining lymph nodes (SDLN) (axillary, brachial, and inguinal) were processed as described previously (22) before staining for flow cytometry.



## Flow cytometry

Single-cell suspensions were initially Fc-blocked using anti-CD16/anti-CD32 (Bio X Cell) and surface-stained in addition to the fixable viability Ghost Dye™ Violet 510 (Tonbo Biosciences) for 30 min at 4 °C in PBS containing 2% FBS. Intracellular staining was performed in perm/wash buffer at 4 °C overnight after permeabilization of cells using the Cytofix/Cytoperm kit (BD Biosciences). For intracellular cytokine staining, isolated cells were incubated in media containing 1x cell stimulation cocktail (Tonbo) at 37 °C, 5% CO<sub>2</sub> and transferred to 4 °C after 6 h. Surface and intracellular staining was performed as described on the next day. To determine absolute cell numbers, CountBright™ Absolute Counting Beads (Invitrogen) were added before measurement. Antibodies used for flow cytometry are summarized in Supplementary Table S1.

Measurement of stained cells was performed on a BD Fortessa flow cytometer using FACS Diva software (BD Biosciences). FlowJo version 10.8.1 was used to analyze cell populations as depicted in supplementary Figure S1. As no markers to distinguish NK cells from ILC1s were included, CD3-NK1.1+ cells are defined as “NK-like” ILC1s and further referred to as ILC1s.

## Histology

Skin strips (approx. 4–5 mm width, 10–20 mm length) were fixed in 10% neutral-buffered formalin and paraffin-embedded for histological analysis. Hematoxylin and eosin (H&E) staining of tissue was performed by University of California, San Francisco Dermatopathology Service and visualized with an Aperio AT2 scanner (Leica Biosystems) at 40x resolution using a 20×0.75NA Plan Apo objective with a 2x optical magnification changer. Image resolution was 40x: 0.25 µm/pixel. Aperio ImageScope software (version 12.4.0.5043, Leica Biosystems) was used to digitally view slides. Pathological evaluation was performed by a board-certified veterinary pathologist.

## Statistics

Statistical analysis was performed using GraphPad Prism (version 9) and paired t-tests were applied to determine *p*-values as indicated in the figure legends. In case of Rosa26<sup>CreERT2</sup>xCD25<sup>fl/fl</sup> mice, animals with low reduction of CD25 (<30% reduction on CD45+ cells after 4-OHT application and IL-2 injection as compared to control side) were excluded from analysis. For all other experiments no animals were excluded.

## Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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## Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## List of abbreviations

|                                |                                 |
|--------------------------------|---------------------------------|
| <b>4-OHT</b>                   | 4-Hydroxytamoxifen              |
| <b>B6</b>                      | C57BL/6 mice                    |
| <b>DC</b>                      | dendritic cell                  |
| <b>DETCs</b>                   | dendritic epidermal T cells     |
| <b>IFN-<math>\gamma</math></b> | interferon- $\gamma$            |
| <b>IL-2</b>                    | interleukin-2                   |
| <b>IL-2R</b>                   | interleukin-2 receptor          |
| <b>ILC1</b>                    | group 1 innate lymphoid cell    |
| <b>ILC2</b>                    | group 2 innate lymphoid cell    |
| <b>MFI</b>                     | mean fluorescence intensity     |
| <b>NK</b>                      | natural killer                  |
| <b>SDLN</b>                    | skin-draining lymph nodes       |
| <b>Teffs</b>                   | T effector cells                |
| <b>TNF-<math>\alpha</math></b> | tumor necrosis factor- $\alpha$ |
| <b>Tregs</b>                   | regulatory T cells              |

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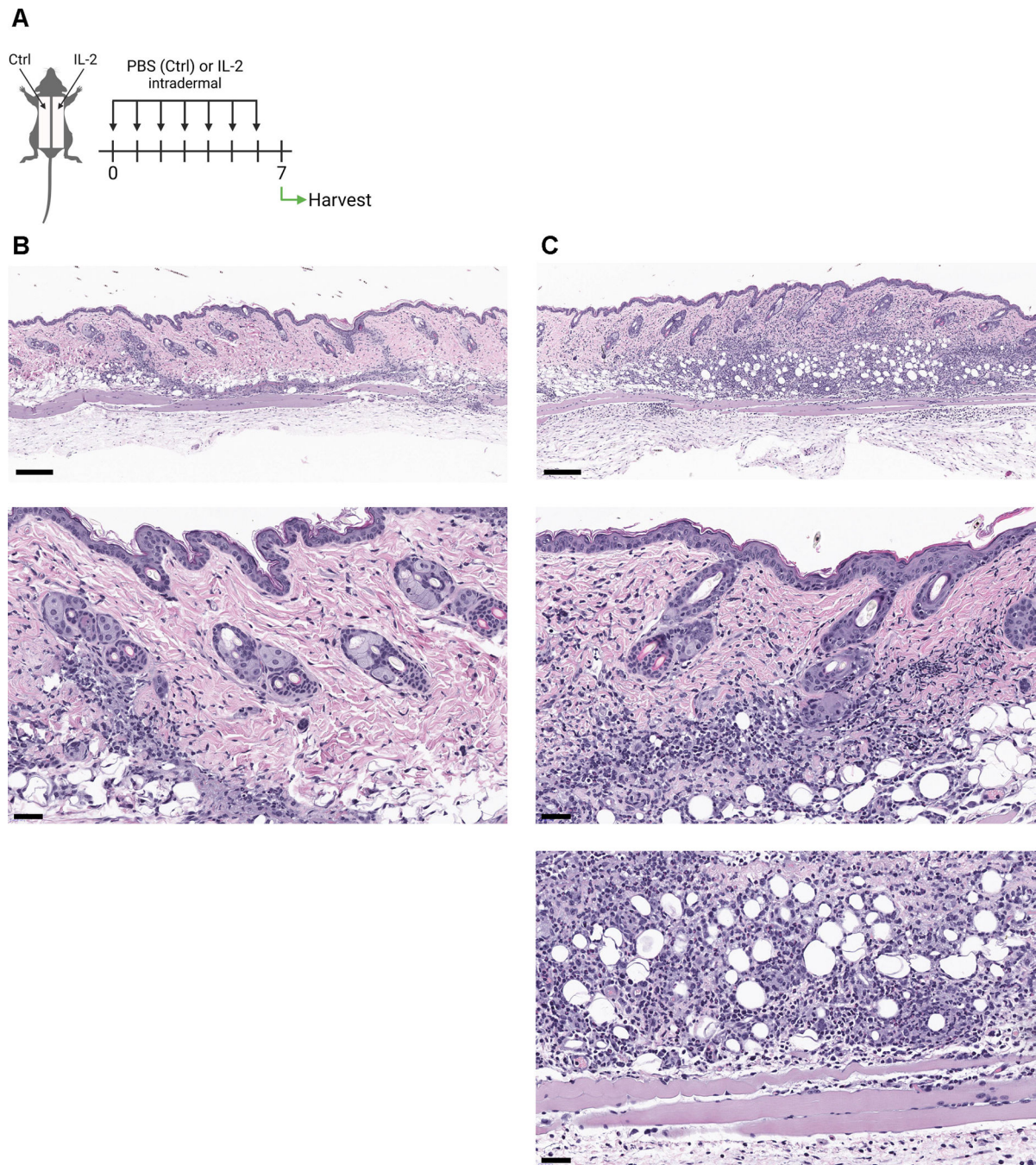
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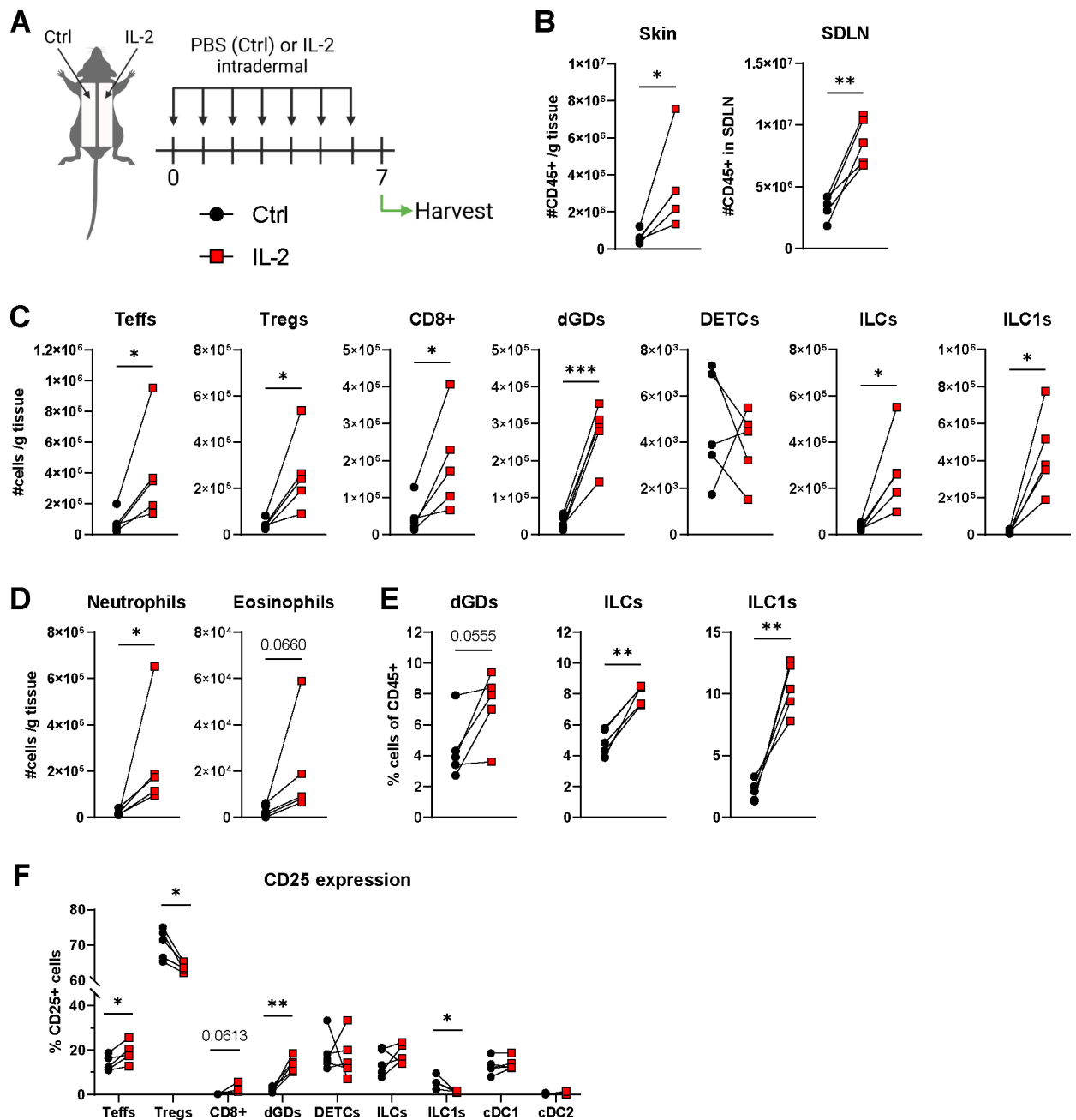




**Figure 1: Intradermal IL-2 injections induce skin inflammation characterized by considerable immune cell infiltrates.**

**A** Wild type mice were intradermally injected with IL-2 (aldesleukin) daily for 7 consecutive days before analysis of skin. Created with [BioRender.com](https://www.biorender.com). Representative histological images of injections sides from control (**B**) and IL-2-treated skin (**C**). Overview pictures (upper images; scale bars 200  $\mu$ m) and close-ups (middle and lower images; scale bars 40  $\mu$ m) are shown.





**Figure 2: IL-2-induced skin inflammation is characterized by increases in frequencies of innate immune cells.**

A Wild type mice were intradermally injected with IL-2 (aldesleukin) daily for 7 consecutive days before analysis of skin and skin-draining lymph nodes (SDLN). Created with [BioRender.com](https://BioRender.com). B Absolute CD45+ immune cell numbers in skin and SDLN. C Absolute numbers of T cell subsets (Teffs: CD4+ T effector cells; dGDs: dermal  $\gamma\delta$  T cells; DETCs: dendritic epidermal T cells), innate lymphoid cells (ILCs), and CD3-NK1.1+ group 1 ILCs (ILC1s). D Absolute numbers of neutrophils and eosinophils in skin. E Frequencies of innate immune cells upon IL-2 injection. F CD25 expression as shown by CD25+ frequencies of the respective cells. cDC: conventional dendritic cell. n=5 mice from

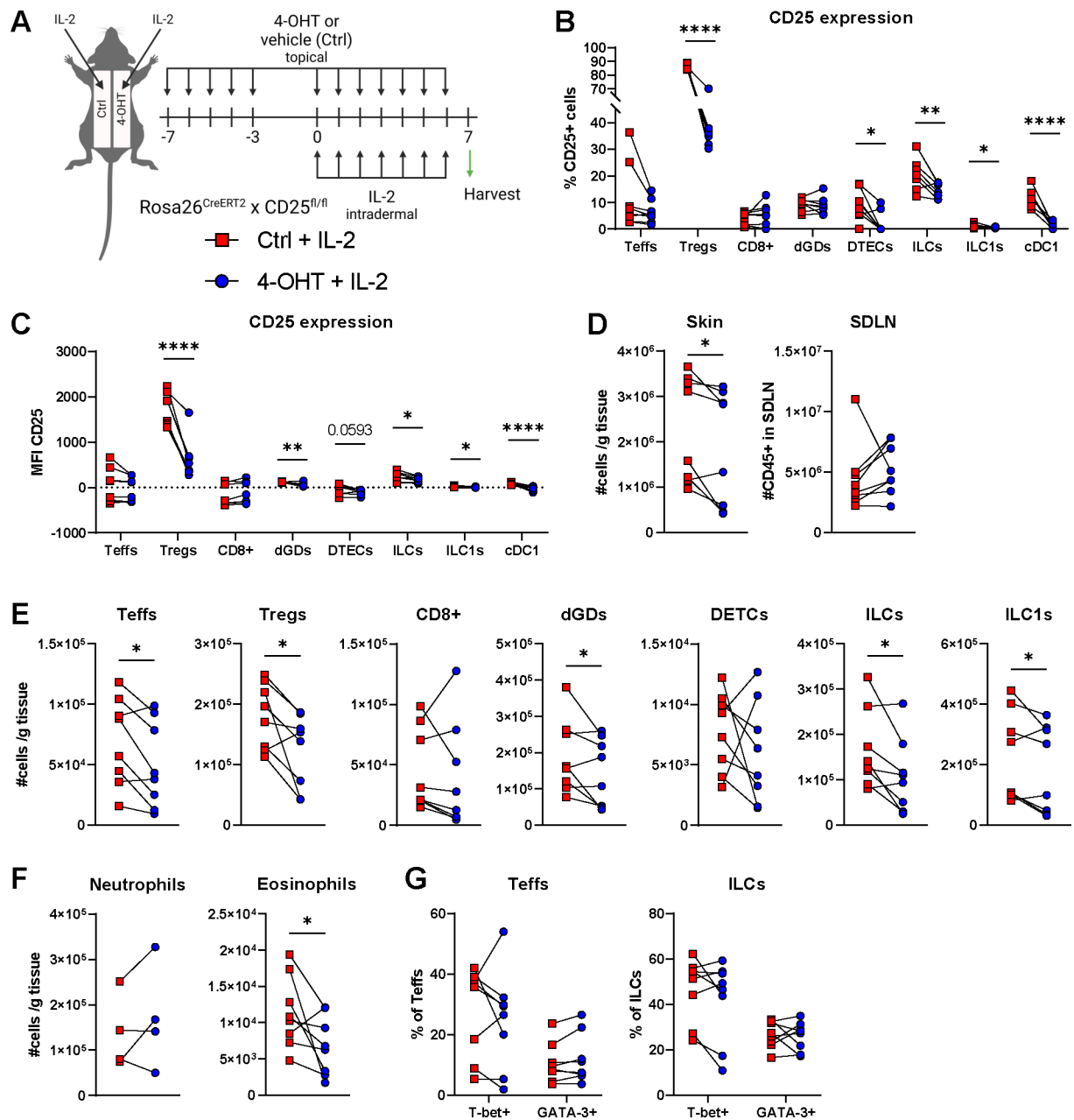
two independent experiments; \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$  as determined by paired t-tests. All other conditions n.s.

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**Figure 3: Reduction of CD25 expression on skin cells reduces IL-2-induced increases of skin immune cell numbers.**

A Cre recombinase of Rosa26<sup>CreERT2</sup>xCD25<sup>fl/fl</sup> mice was induced by topical 4-Hydroxytamoxifen (4-OHT) application prior to IL-2 injections on both sides to decrease CD25 on all skin cells. Acetone on the contralateral side of the back of the same mouse served as vehicle control. Created with [BioRender.com](https://www.biorender.com). **B** CD25 expression as shown by CD25+ cell frequencies of the respective cell subsets. Teffs: CD4+ T effector cells; dGDs: dermal  $\gamma\delta$  T cells; DETCs: dendritic epidermal T cells; ILCs: innate lymphoid cells; ILC1s: CD3-NK1.1+ group 1 ILCs; cDC1: type 1 conventional dendritic cell. **C** Mean fluorescence intensity (MFI) of CD25 on respective cells. **D** Immune cell numbers on control and 4-OHT-treated sides in skin and skin-draining lymph nodes (SDLN). **E** Absolute cell numbers of

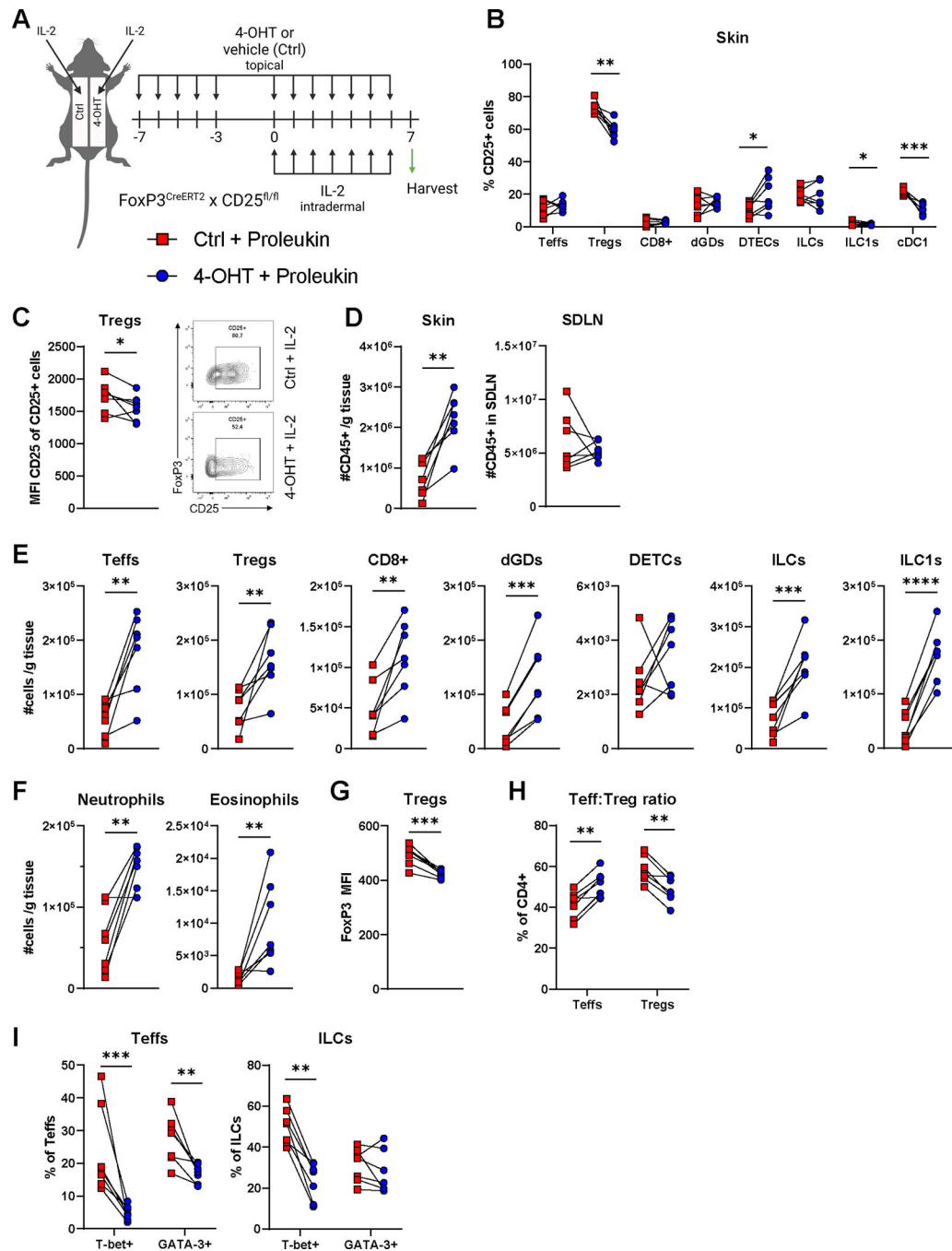
lymphocyte immune cell subsets in skin tissue. **F** Absolute numbers of neutrophils and eosinophils in skin. **G** Expression of T-bet at GATA-3 transcription factors in Teffs and ILCs. n=4–8 mice from three independent experiments; \*  $p<0.05$ , \*\*  $p<0.01$ , \*\*\*\*  $p<0.0001$  as determined by paired t-tests. All other conditions n.s.

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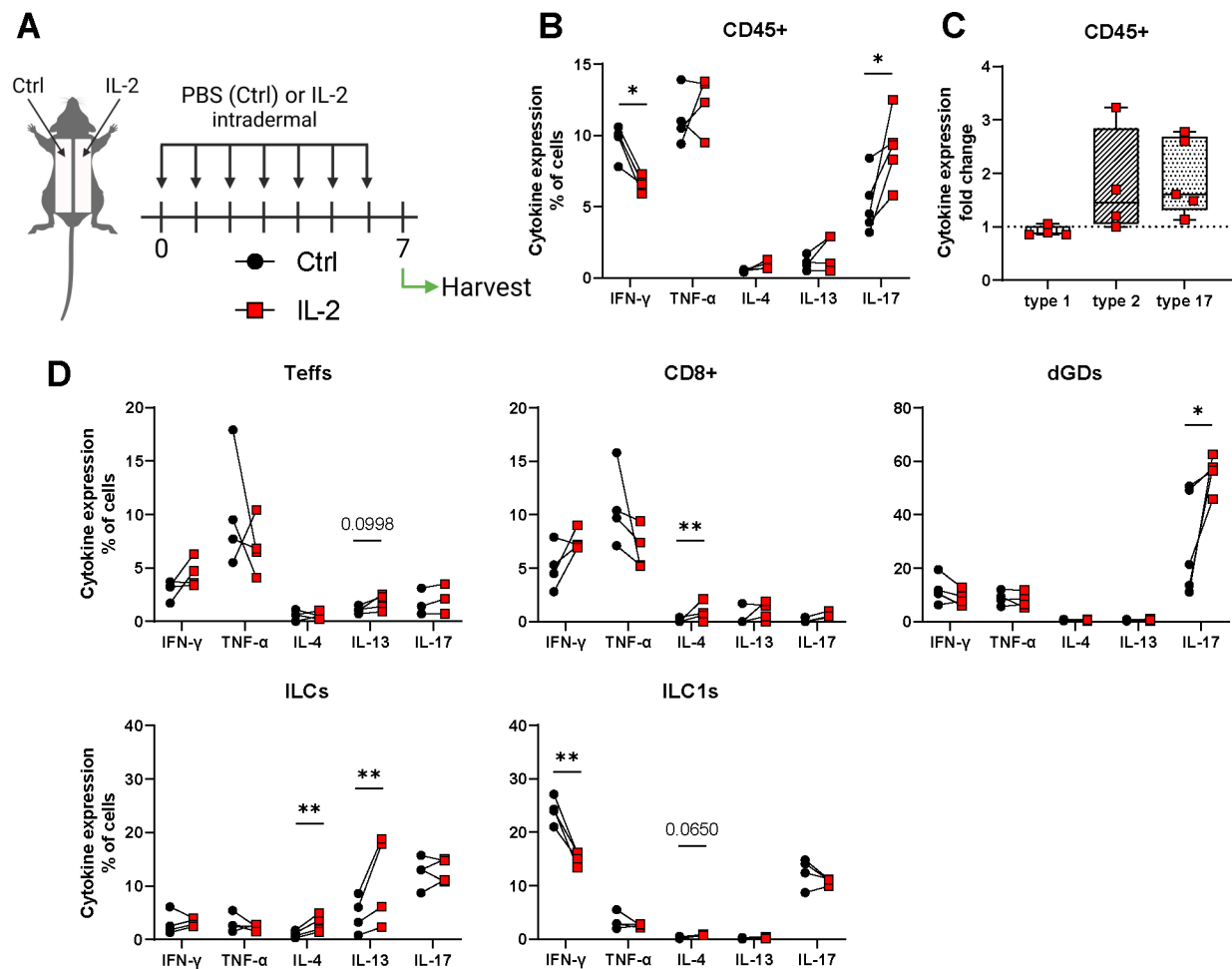


**Figure 4: Reduction of CD25 on Tregs leads to significant increases in IL-2-induced skin inflammation.**

**A** Cre recombinase of FoxP3<sup>CreERT2</sup>xCD25<sup>fl/fl</sup> mice was induced by topical 4-Hydroxytamoxifen (4-OHT) application to delete CD25 on skin Tregs prior to IL-2 injections on both sides. Acetone on the contralateral side of the back of the same mouse served as vehicle control. Created with [BioRender.com](https://www.biorender.com). **B** CD25 expression as shown by CD25<sup>+</sup> frequencies of the respective cells. Teffs: CD4<sup>+</sup> T effector cells; dGDS: dermal  $\gamma\delta$  T cells; DETCs: dendritic epidermal T cells; ILCs: innate lymphoid cells; ILC1s:



CD3-NK1.1+ group 1 ILCs; cDC1: type 1 conventional DC. **C** CD25 mean fluorescence intensity (MFI) of CD25+ Tregs (left) and representative CD25 gating on Tregs (right). **D** Immune cell numbers on control and 4-OHT-treated sides in skin and skin-draining lymph nodes (SDLN). **E** Absolute cell numbers of immune cell subsets in skin tissue. **F** Absolute numbers of neutrophils and eosinophils in skin. **G** MFI of FoxP3 in Tregs in skin. **H** Frequencies of skin Teffs and Tregs of CD4+ cells. **I** Expression of T-bet at GATA-3 transcription factors in Teffs and ILCs. n=7 mice; \*  $p<0.05$ , \*\*  $p<0.01$ , \*\*\*  $p<0.001$  as determined by paired t-tests. All other conditions n.s.



**Figure 5: IL-2 injections induce a mixed type2/type 17 immune reaction in skin, associated with dermal  $\gamma\delta$  T cells and ILCs.**

**A** Experimental set-up for injection of wild type mice with IL-2. Created with [BioRender.com](https://www.biorender.com). **B** Expression of cytokines in CD45+ cells upon IL-2 injection. **C** Relative cytokine expression in immune cells in IL-2-treated vs. Ctrl skin. Frequencies of cytokine expression in CD45+ cells (type 1: IFN- $\gamma$  + TNF- $\alpha$ ; type 2: IL-4 + IL-13; type 17: IL-17) upon IL-2 injection was normalized to control conditions to yield fold changes. **D** Cytokine expression in different lymphocyte populations. Teffs: CD4+ T effector cells; dGDs: dermal  $\gamma\delta$  T cells; ILCs: innate lymphoid cells; ILC1s: CD3-NK1.1+ group 1 ILCs. n=4–5 mice representative for two independent experiments; \*  $p < 0.05$ , \*\*  $p < 0.01$  as determined by paired t-tests. All other conditions n.s.